Ex.No: 1
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Study of various Network commands used in Linux and Windows

Aim:

To study various network commands used in Linux and Windows

Commands:

1. arp -a : Address Resolution Protocol, It show IP address along with IP and MAC address of the router.
2. hostname : It displays the name of computer
3. ipconfig : It displays detailed config information about PC P/IP connection
4. nbtstat -a: solves problem with NETBIOS name resolution
5. netstat : displays variety of network stats
6. nslookup: It is a tool for DNS lookup, returns IP address and MX, NS records
7. pathping : It traces the route the packet send to
8. ping : tests connectivity btw two nodes
9. route : used to show/ manipulate IP routing table.

Linux commands:

1. ip : basic command used to show all the interfaces. Adds IP to interface
2. ifconfig : used for troubleshooting the network
3. mtr : It traces the route the packet send to.
4. tcpdump: helps to monitor the tcp packets Send and recieving
5. ping : same as windows.
6. nmcli : If you are connected over ethernet, you can use nmcli to configure.

Examples:- (Windows)

1. arp -a:

Interface: 192.168.1.5

Internet address	Physical Address	Type
192.168.1.1	00-11-22-44-55	dynamic
192.168.1.20	aa-bb-cc-ff-gg	dynamic

2. hostname:

Office - Desktop - PC

3. ipconfig/all:

Ethernet adapter Ethernet 0:

Description.....: Intel(R) Ethernet
Physical address.....: 00-1A-2B-3C-4D
DNS server.....: 8.8.8.8
DHCP enabled.....: Yes

4. nbstat -a

NetBIOS Remote Machine Name Table

Name	Type	Status
Remoteserver <00>	Unique	Registered
Workgroup <00>	group	Registered

MAC - address = 00-5a-56-A1-B2

5. netstat -r

IPV4 Route table

Active Routes:

	Network	Gateway
0.0.0.0	0.0.0.0	192.168.1.1

6. nslookup:

Server : 8.8.8.8
Address : 8.8.8.8

Non - authoritative answer:
Name : www.google.com
Address : 142.250.182.68

7. pathping:

Tracing route to www.google.com
0 My-PC
1 192.168.1.1
2 ISP Router
3 Google Server
Computing statistics for 100 seconds...

8. ping:

Pinging localhost [127.0.0.1]
Reply from 127.0.0.1: bytes = 32 time <1ms TTL=128
Reply from 127.0.0.1: bytes = 32 time <1ms TTL=128

9. route:

Kernel IP routing table

Destination	Gateway	Genmask	Flags
0.0.0.0	192.168.1.1	0.0.0.0	UG
192.168.1.1	0.0.0.0	255.255.0	U

10. nmcli:

Device	Type	State	Connection
eth0	ethernet	connected	Wired connection 1
lo	wifi	disconnected	--

Examples:- (Linux)

1. ip:

```
1: lo: <LOOPBACK, UP, LOWER_UP> mtu 65536
    inet 127.0.0.1/8 scope host lo

2: eth0: <BROADCAST, MULTICAST, UP> mtu 1500
    inet 192.168.1.100/24 brd scope global eth0
```

2. ifconfig:

```
eth0: flags=4163<UP, BROADCAST, RUNNING>
      inet      192.168.1.100
      netmask   255.255.0.0
      broadcast 192.168.1.255

lo: flags=73<UP, LOOPBACK, RUNNING>
      inet 127.0.0.1
```

3. mtr:

	Host: localhost	Loss%	Snt	Last	Avg	Best
1.	192.168.1.1	0.0%	10	1.2	1.3	1.0
2.	10.20.0.1	0.0%	10	5.4	5.8	5.1

4. tcpdump:

```
12:30:10 IP 192.168.1.1 > 8.8.8.8.53 : UDP
12:30:10 IP 8.8.8.8.53 > 192.168.1.1 : UDP
```

5. ping:

```
PING google.com (142.250.183.73)

64 bytes from 142.250.183.73 icmp_seq=1 time=18
64 bytes from 142.250.183.73 icmp_seq=2 time=17
```

Result:

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Thus the basic commands for Linux and windows are studied.